MIL-C-13212B (AR)
Amendment 2
6 October 1981
SUPERSEDING
Amendment 1
17 January 1969

MILITARY SPECIFICATION

COMPONENTS, MOLDED, ASPHALT-ASBESTOS FILLED (FOR USE IN AMMUNITION SHIPPING CONTAINERS)

This amendment forms a part of Military Specification MIL-C-13212B (MU), dated 15 November 1968, and is approved by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

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* Add the following preamble underneath the title:

"This specification is approved by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense."

* 1.1, Scope: Delete in its entirety and substitute the following:

"1.1 Scope. This specification covers components molded of one grade of asbestos filled asphaltic molding material, with or without metallic inserts, and used in shipping containers for ammunition (see 6.1)."

1.2.1, line 1: Delete "components" and substitute "molding material."

* 2, Applicable Documents, Standards, Federal: Delete "FED-STD-151 - Metals, Test Methods."

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* Add new paragraph 2.2 as follows:

"2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARD:

B117-64 -Salt Spray (Fog) Testing

D792-66 -Specific Gravity and Density of Plastics by Displacement.

D224068 -Indentation Hardness of Rubber and Plastics by means of a Durameter, Test For."

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103)".

* 3.1: Delete in its entirety and substitute the following:

"3.1 Material. Components covered by this specification shall be made of asbestos-filled asphaltic material conforming to MIL-P-43090, hereinafter referred to as "plastic", with or without metallic inserts. Unless otherwise specified, the color of the molding compound shall be black. The metallic inserts, when used, shall be of the composition specified on the drawing or specification covering the particular component."

3.4, line 1 and 5: Delete "Molded components" and substitute "Components after molding."

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* 3.5, line 1: Delete "molded" and substitute "plastic portion of the."

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* Add new paragraph 3.8 as follows:

"3.8 Inorganic residue. The plastic of the component shall be capable of producing an inorganic residue equal to 20.0 ± 7.5 percent or the weight of the specimen when tested as specified in 4.4.4."

* Add new paragraph 3.9 as follows:

"3.9 Durometer hardness. The plastic portion of the component shall meet the following durometer hardness when tested as specified in 4.4.5:

$23^{o} \pm 1^{o}C$ ($73.4^{o} \pm 1.8^{o}F$) - 71 ± 6 (D/65 - 77/1)

$71^{o} \pm 2^{o}C$ ($160^{o} \pm 3.6^{o}F$) - 59 ± 6 (D/53 - 65/1)."

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* Add new paragraph 4.1.2 as follows:

"4.1.2 Submission of test specimens. In addition to the submission of the test components specified, a complete set of test specimens from each lot will be prepared and furnished an approved laboratory, together with ten complete assemblies for testing. The specimens shall be molded by the contractor from the material used to mold the components. The required specimens shall be prepared in the forms and quantities suitable for testing for compliance with Specification MIL-P-43090."

4.2.1, line 8: After "contract." add the following sentence "In addition to the components specified, ten test specimens conforming to the requirements of MIL-P-43090 shall be furnished."

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4.2.2, line 1: Delete "and components" and substitute ", components and test specimens."

4.2.3:, line 1: Delete "or component" and substitute ", component or test specimen."

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4.3.3.1:

Line 3: Add "(See 6.5)." after "Level S-4"

Defect (1): Delete "drawing" and substitute "drawings 8799117 and 9216180)."

4.3.3.2:

Major defect 104: Delete in its entirety.

Minor Category: Delete "AQL 2.5 percent" and substitute "AQL 1.5 percent."

Minor defect 202: Delete in its entirety.

Minor defect 204: Delete in its entirety.

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Add new paragraph 4.3.3.5 as follows:

"4.3.3.5 Support assembly, prior to application of asphalt-asbestos material (see dwg. 9216180)."

Categories	Defects	Method of Inspection	Code No.
Critical:	Node defined.		
Major:	None defined.		
Minor:	AQL 0.65 percent		
	201. Crimp of metal components incorrect	Visual-	02008"

* 4.3.4.1: Delete in its entirety and substitute the following:

"4.3.4.1 Specify gravity (see 3.5), Major defect, Code No. 05001; Inorganic residue (see 3.d), Major defect, Code No. 06001; Durometer hardness (see 3.9), Major defect, Code No. 07001. The sampling plans for these tests shall be in accordance with MIL-STD-105, using inspection level S-2 and an AQL of 2.5 percent. Failure to meet the applicable requirement when tested as specified in 4.4 shall be cause for rejection of the lot."

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Add new paragraphs 4.3.4.2 and 4.3.4.3 as follows:

"4.3.4.2 Core support missing, Major defect, Code No. 02003; Location of core support from forward face, Minor defect, Code No. 02006 (see 6.5). Ten support assemblies shall be selected from each lot for this examination. Failure of any assembly to meet the applicable requirements shall be cause for rejection of the lot. All samples subjected to the tests shall be classed defective and removed from the lot. The tests shall be performed as specified in 4.4.2, using test equipment in accordance with 4.3.5."

"4.3.4.3 Salt spray test, Major defect, Code No. 10001. Sampling for this test shall be in accordance with Federal Standard, FED-STD-151. Failure of one or more assemblies to meet the applicable requirement shall be cause for rejection of the lot."

4.3.5: Delete in its entirety and substitute the following:

"4.3.5 Inspection equipment. For the performance of all tests and examinations specified in 4.3 and 4.4, commercial inspection equipment should be employed (see 6.4). The contractor shall have available, and utilize correctly, this equipment and is charged with the responsibility of insuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes."

4.4.1, line 3: Delete "5011 of FED-STD-406." and substitute "A1 of ASTM D-792-66."

Add new paragraphs 4.4.2 and 4.4.3 4.4.4, 4.4.5 as follows:

"4.4.2 Core support missing; Location of core support from forward face. The asphalt-asbestos molding material covering the assembly shall be removed from the core support in two areas 90 degrees apart, by any practical method. Care should be exercised in order to prevent deformation or chipping of the material at the forward face. Upon removal of the molding material, examinations shall be performed utilizing the following methods:

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Defect	Method of Inspection
Location of support from forward face	Gage
Core support missing	Visual

* 4.4.3 Salt spray test. The support assemblies shall be tested as specified in ASTM-B117-64.

* 4.4.4 Inorganic residue. Pieces of sufficient size plastic shall be cut or broken from the component and tested in accordance with the inorganic residue test method specified in MIL-P-43090. The percent inorganic residue shall be calculated and reported.

* 4.4.5 Durometer hardness. The tests shall be performed in accordance with ASTM D2240-64T, with the following exceptions and additions: (1)

a. A complete component after molding shall be used as the specimen.

b. The type D durometer shall be pressed into the specimen at points where the thickness, area, curvature, and smoothness of the plastic layer meets the requirements specified for the specimen in the method. At least three readings shall be taken on each specimen, and the average results shall be reported.

c. After the specimens have been tested at standard room temperature, they shall be heated for one hour at $71^{\circ} \pm 2^{\circ}C$ ($160^{\circ} \pm 3.6^{\circ}F$) and retested within one minute after removal from the oven. The average for each specimen, at each temperature, shall conform to the applicable requirements of 3.9.

(1) This method applies only to components which conform to the thickness, area, curvature, and smoothness properties of ASTM D2240-64T."

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Add new paragraphs 6.4, 6.4.1, 6.4.2, 6.5, and 6.6 as follows:

"6.4 Inspection equipment

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6.4.1 Commercial inspection equipment is defined in AMC Regulation 702-2.

6.4.2 Contractor-designed equipment. In the event that a contractor elects to design his own inspection equipment, details of the designs (drawings, description, materials, etc.) shall be submitted to: Commander, USA ARRADCOM, Dover, New Jersey 07801, ATTN: DRDAR-QAR-I, for approval prior to fabrication and use. Approval of such designs may be delegated to the contract administration office for minor defects only.

6.5 Verification of composition. Test specimens representative of the material used in each production lot, together with the 10 assemblies subjected to the tests specified in 4.3.4.2 shall be furnished an approved Government laboratory for verification of composition. The test specimens shall be furnished simultaneously with the shipment of each production lot and shall be sufficient in number to permit testing for conformance to all requirements of Specification MIL-P-43090. From the 10 assemblies, test samples suitable for determination of inorganic residue and shore hardness shall be cut. Variations in fiber length, distribution, and nature of the asbestos reinforcement and brittleness of the asphalt shall be determined by breaking the material from the assembly and noting the resistance to breaking and the visual appearance of the fractures. Additionally, burning the asphalt from both specimens and assemblies and comparing the visual appearance of the inorganic residues shall be performed. Any significant difference between the test specimens and assemblies shall be considered sufficient justification for rejection of the lot."

6.6 Submission of test data. In addition to the normal distribution of records, two copies of the test data shall be furnished: Commander, USA ARRADCOM, Dover, New Jersey, ATTN: DRDAR-QAR-Q.

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